# COMPUTER NETWORKS
# PHASE 1 – LITERATURE REVIEW AND RESEARCH GAP IDENTIFICATION

**Proposed Research Topic:**

Adaptive Network Intrusion Detection and Early Warning through Game-Theoretic Scenario Generation and Temporal Network Analysis

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## Critical Summary

### Paper 1 – Deepanramkumar and Sharmila, "GraphINR-IDS: Graph-Topology-Aware Implicit Neural Reconstruction for Zero-Day Intrusion Detection in Encrypted Network Traffic." [1]

The authors study intrusion detection using only benign network-flow observations for fitting, an approach intended to reduce dependence on enumerating known attacks. Their continuous implicit neural representation combines communication-graph Laplacian coordinates, multi-frequency temporal encoding and a SIREN reconstruction network. They evaluate UNSW-NB15 and CIC-IDS2017 under random and chronological protocols, reporting F1 $99.06 \pm 0.12\%$, AUROC 99.66%, AUPRC 99.88% and false-positive rate 5.13% in the specified primary setting. The paper explicitly identifies calibration drift under chronological transfer, unseen-endpoint handling and the need for deployment studies as challenges. Further limitations inferred here are that offline classification is not a demonstrated *warning before an incident milestone*, and the datasets do not demonstrate outcomes of strategically selected live network episodes. Its official implementation and both underlying datasets are openly referenced; CIC-IDS2017's supplied CSV representation omits the full identifiers/timestamps necessary for some graph-temporal tests.

### Paper 2 – Bala et al., "Temporal Flow-Based Dynamic Graph Neural Network with Robustness Interpretation for DoS/DDoS IoT Intrusion Detection." [2]

This paper represents ToN-IoT communication flows as directed temporal graphs, with host relations and engineered measures including source dispersion and traffic concentration. It tests a GATv2 detector with feature selection, post-hoc explanation and projected-gradient adversarial training against DoS, DDoS and normal traffic. Reported test balanced accuracy is 0.9996 and macro-F1 0.9979 under the stated experiment; robustness tests show performance loss under perturbations despite hardening, and a natural-prevalence evaluation is materially less favourable. Author-discussed weaknesses and our assessment include sensitivity to class mixture, flow-feature perturbations that need not reproduce physical network episodes, a three-class threat scope, and dependence on the particular graph/split construction. This establishes useful evidence for temporal graph classification but *not* that a detector issues a causally valid alert ahead of subsequent lateral movement or data transfer. The authors publish their analysis notebooks and identify ToN-IoT as the data source.

### Paper 3 – Afrifah et al., "Multi-stage Attack Detection in IIoT Supply Chains Using Graph Neural Networks." [3]

The study maps flows to four operational stages—benign, initial access/discovery, lateral movement/execution/persistence, and impact—then compares Random Forest, GCN, GAT, GraphSAGE and fusion on ToN-IoT and Edge-IIoTset. Its corrected, asset-aware ToN-IoT pipeline reports GAT macro-F1 $0.687 \pm 0.041$, whereas on Edge-IIoTset a tabular Random Forest is strongest (macro-F1 0.615). Most importantly, the authors explicitly decline to claim pre-impact forecasting: among 21,313 candidate pre-impact records, only five positives exist, all in training, leaving none for validation or held-out evaluation. This is an unusually direct illustration of why *stage classification* and validated *early warning* are different tasks. Other stated constraints include imperfect timestamps in one benchmark and concentration of malicious events in few assets; cross-network topology transfer and strategic episode generation remain untested. A paper-linked code address and public dataset sources are given, but execution of that code has not been independently verified here.

## Cross-Paper Comparison Table

| # | Title | Year | Dataset Used | Preprocessing Techniques | Models / Techniques Used | Evaluation Metrics | Results | Limitations / Research Gap |
|---|---|---|---|---|---|---|---|---|
| 1 | GraphINR-IDS: reconstruction | 2026 | UNSW-NB15; CIC-IDS2017 | Benign-only fit and calibration; graph spectral coordinates; continuous-time features | SIREN-based flow reconstruction; one-class residual score | F1; AUROC; AUPRC; FPR; chronological transfer; ablations | Primary protocol F1 99.06 ± 0.12%; FPR 5.13% | Calibration drift; static/limited topology; no staged network warning or scenario policy |
| 2 | Temporal GATv2: DoS/DDoS | 2026 | ToN-IoT traffic subset | Directed temporal flow graphs; entropy, statistical and selection features; natural-class evaluation | GATv2; PGD adversarial training; GNNExplainer; graph baselines | Balanced accuracy; macro-F1; MCC; robustness versus perturbation | Balanced accuracy 0.9996; macro-F1 0.9979 in specified test | Three labels; perturbation is feature-based; cross-topology staged warning not evaluated |
| 3 | Multi-stage IIoT detection | 2026 | ToN-IoT; Edge-IIoTset | Asset-disjoint split; time-window graphs; stage mapping; data-error correction | RF; GCN; GAT; GraphSAGE; fusion; prospective pre-impact protocol | Macro-F1; sensitivity; bootstrap; inference latency; forecasting feasibility | ToN-IoT GAT 0.687 ± 0.041 macro-F1; Edge-IIoTset RF 0.615 | No held-out pre-impact positives; identity concentration; differing timestamp quality |

**Comparative interpretation.** Paper 1 asks whether benign-only reconstruction can identify novel traffic; Paper 2 asks whether temporal flow graphs remain classifiable under adversarial feature perturbations; Paper 3 asks whether network behaviour can be assigned interpretable attack stages. Their different labels and testing protocols make their headline F1 values *not directly comparable*. None establishes a budget-controlled, repeated attacker–defender *network experiment* that measures how much *valid early-warning lead time* results from strategic rather than fixed or random scenario selection.

## Research Gap Synthesis

| Research Gap | Evidence: Paper 1 | Evidence: Paper 2 | Evidence: Paper 3 | Proposed Research Opportunity |
|---|---|---|---|---|
| **Gap 1: Verified pre-impact warning** | Scores present-flow anomalies; no future milestone labels. | Evaluates attacks and perturbations, not alerts before a separately logged milestone. | Pre-impact task defined but no positive held-out examples. | Produce labelled, time-stamped multi-stage episodes; measure valid $T_{\text{lead}}$, warning coverage and false alarms. |
| **Gap 2: Strategic scenario selection** | Uses observational benchmark splits, not an adaptive episode generator. | Uses crafted feature perturbations, not matched-budget attacker–defender episodes. | Stage-aware detection without a repeated scenario-selection game. | Compare fixed, uniform-random and finite-game selection with identical episode budgets and independent tests. |
| **Gap 3: Generalisation to unfamiliar communication structure** | Chronological and endpoint limits remain; graph conditions are constrained. | ToN-IoT host-flow graphs and narrow DoS/DDoS scope. | Dataset changes cause different best models; malicious assets are sparse. | Reserve whole episode families and, if enough data exist, complete network topologies for testing. |
| **Gap 4: Scientific validity of "zero-day-like" behaviour** | Benign-only fitting excludes labelled attacks but does not prove authentic zero-day detection. | Adversarial perturbations are not new exploits or verified unseen propagation. | Stage mapping and missing longitudinal evidence constrain future-event claims. | Report previously unseen *behavioural profiles* only; keep exploit provenance, milestone ground truth and uncertainty explicit. |

**Proposed research question.** *Can game-theoretic adversarial scenario generation improve detection and causally valid early warning of previously unseen multi-stage network behaviours, compared with fixed and uniformly sampled scenarios, under matched data budgets and independently held-out network conditions?*

**Testable distinction.** A valid warning is raised only after an independently defined onset and before a later, independently logged network milestone (for example, controlled movement between service segments or transfer of harmless test data). Merely detecting the stage in which impact is already occurring is not early warning.

## References

[1] P. Deepanramkumar and A. Helen Sharmila, “GraphINR-IDS: Graph-Topology-Aware implicit neural reconstruction for zero-day intrusion detection in encrypted network traffic,” *Journal of King Saud University – Computer and Information Sciences*, 2026. DOI: `10.1007/s44443-026-01260-x`.

[2] T. J. Bala et al., “Temporal flow-based dynamic graph neural network with robustness interpretation for DoS/DDoS IoT intrusion detection,” *Defence Technology*, 2026. DOI: `10.1016/j.dt.2026.06.002`.

[3] W. Afrifah, G. Epiphaniou, H. S. Lallie, and C. Maple, “Multi-stage attack detection in IIoT supply chains using graph neural networks,” *International Journal of Information Security*, vol. 25, art. 176, 2026. DOI: `10.1007/s10207-026-01340-z`.

## Paper Selection and Reproducibility Check

| **Study** | **Journal rank (2025)** | **Original dataset** | **Research code / access status** |
|---|---|---|---|
| P1 | SJR Q1; JCR Q1 | UNSW-NB15; CIC-IDS2017 | `https://github.com/Deepan777/GraphINR-IDS` – public repository indexed; not re-executed here. |
| P2 | SJR Q1; JCR Q1 (engineering category) | ToN-IoT | `https://github.com/jencyyyyy/TF-GATv2` – public notebooks indexed; dataset checkpoints on linked Kaggle pages. |
| P3 | SJR Q1; JCR Q1 | ToN-IoT; Edge-IIoTset | `https://github.com/AfrifaW/Multi-Stage-Attack-Detection` – author-declared repository; independent availability/execution not established. |

**Rank and evidence note.** Journal quartiles reflect the 2025 SJR/JCR editions, not a presumed 2026 ranking. They were checked against journal-index records (including the ISSN-matched profiles at `https://www.akaturk.com/` and `https://scienceaijournal.com/`); formal university assessment should use its prescribed database/category. All reported scores are the cited authors’ findings; no experiments in the proposed project have yet been performed. “Zero-day-like” denotes held-out behavioural novelty, not proven zero-day exploitation.